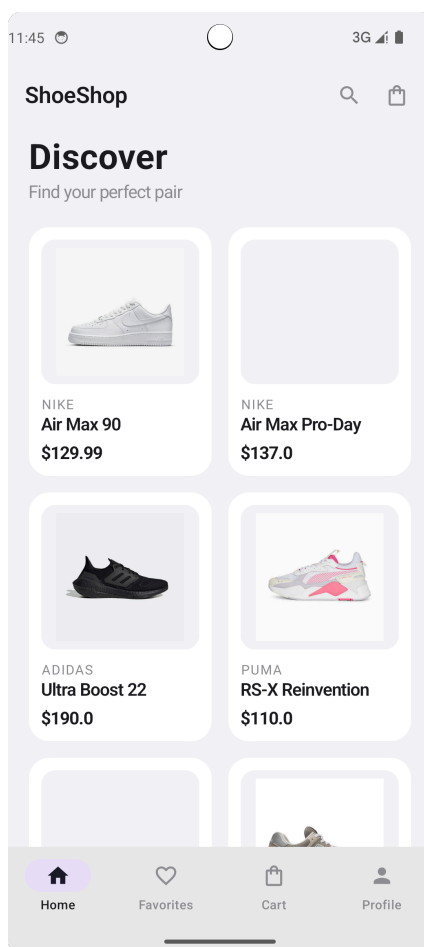


ShoeShop

Kotlin Jetpack Compose Mobile Shopping App



Course: CS5450 Mobile Programming

Group: Group #4 - Mobile Shop

Framework: Kotlin Jetpack Compose (Material3)

Platform: Android (API 26+)

Date: May 5, 2026

Contents

1	Project Overview	2
1.1	Key Architecture Principles	2
2	GitHub Public Repository	2
3	How to Configure and Run	2
3.1	Prerequisites	2
3.2	Steps to Run	2
3.3	Build Configuration	3
4	Project Structure	3
5	Technologies Used	4
6	Architecture Details	4
6.1	ViewModel + StateFlow (State Management)	4
6.2	Type-Safe Navigation (Gold Standard)	4
6.3	State Hoisting & Unidirectional Data Flow	4
6.4	Material Design 3	5
7	Features	5
7.1	Home Screen	5
7.2	Detail Screen	5
7.3	Cart Screen	5
8	Screenshots	6
8.1	Home Screen	6
8.2	Detail Screen	7
8.3	Shopping Cart	8
8.4	Cart with Multiple Items	9

1 Project Overview

ShoeShop is a mobile shopping application built with **Kotlin Jetpack Compose** that demonstrates modern Android development (MAD) practices. The app allows users to:

- Browse a catalog of shoes displayed in a responsive grid layout
- View detailed product information including description, price, and available sizes
- Add items to a shopping bag with quantity management
- View shopping bag total with itemized pricing

1.1 Key Architecture Principles

- **ViewModel + StateFlow** — Cart state managed in ViewModel, survives screen rotation
- **Type-Safe Navigation** — Using @Serializable route objects (Gold Standard)
- **State Hoisting & UDF** — Stateless composables with unidirectional data flow
- **Material Design 3** — Scaffold, TopAppBar, NavigationBar throughout

2 GitHub Public Repository

<https://github.com/realzhengyuting-cloud/ShoeShop>

3 How to Configure and Run

3.1 Prerequisites

- Android Studio Hedgehog (2023.1.1) or later
- JDK 17
- Android SDK 34
- Android Emulator or physical device (API 26+)

3.2 Steps to Run

1. Clone the repository: `git clone https://github.com/realzhengyuting-cloud/ShoeShop.git`
2. Open Android Studio
3. Select **File** > **Open** and choose the **ShoeShop** folder
4. Wait for Gradle sync to complete (first time may take a few minutes)
5. Create a virtual device: **Tools** > **Device Manager** > **Create Virtual Device**

- Select **Pixel 6** (or any phone)
 - Choose system image **API 34**
 - Click Finish
6. Select the emulator from the device dropdown
 7. Click the **Run** button (green play icon) or press **Shift+F10**

3.3 Build Configuration

Component	Version
Kotlin Plugin	1.9.24
Compose Compiler	1.5.14
Navigation Compose	2.8.5
Lifecycle Runtime Compose	2.7.0
Serialization JSON	1.6.3

4 Project Structure

```

ShoeShop/
|-- app/
|   |-- build.gradle.kts          (App-level build config)
|   +-- src/main/
|       |-- AndroidManifest.xml  (App manifest)
|       |-- java/com/example/shoeshop/
|           |-- MainActivity.kt   (Entry point, NavHost)
|           |-- model/
|               |-- Shoe.kt       (Shoe data class)
|               +-- CartItem.kt   (Cart item data class)
|           |-- data/
|               +-- ShoeData.kt   (Static shoe catalog)
|           |-- navigation/
|               +-- Routes.kt     (@Serializable routes)
|           |-- viewmodel/
|               +-- CartViewModel.kt (StateFlow state mgmt)
|           +-- ui/
|               |-- theme/
|                   |-- Color.kt   (Color definitions)
|                   +-- Theme.kt   (Material3 theme)
|               |-- components/
|                   +-- ShoeCard.kt (Stateless card widget)
|               +-- screens/
|                   |-- HomeScreen.kt (Scaffold+NavigationBar)
|                   |-- DetailScreen.kt (Scaffold+TopAppBar)
|                   +-- CartScreen.kt (Scaffold+TopAppBar)
|       +-- res/values/
|           |-- strings.xml

```

```
|          +-- themes.xml
|-- build.gradle.kts          (Root: Kotlin 1.9.24)
|-- settings.gradle.kts      (Project settings)
+-- gradle/wrapper/
    +-- gradle-wrapper.properties(Gradle version)
```

5 Technologies Used

Technology	Purpose
Kotlin 1.9.24	Programming language
Jetpack Compose	Declarative UI framework
Material3	Design system & components
ViewModel	Lifecycle-aware state holder
StateFlow	Reactive state management
Navigation Compose 2.8.5	Type-Safe navigation
Kotlinx Serialization	Route object serialization
Coil	Asynchronous image loading
Compose Animation	Page transitions & interactions

6 Architecture Details

6.1 ViewModel + StateFlow (State Management)

Cart state is managed in `CartViewModel` using `MutableStateFlow`. The UI observes state via `collectAsStateWithLifecycle()` which is the recommended lifecycle-aware approach. This ensures data survives screen rotation (configuration changes).

6.2 Type-Safe Navigation (Gold Standard)

Routes are defined as `@Serializable` objects and data classes:

- `object HomeRoute` — Home screen
- `data class DetailRoute(val shoeId: Int)` — Detail with typed parameter
- `object CartRoute` — Cart screen

Navigation uses object passing: `navController.navigate(DetailRoute(shoeId = shoe.id))`. No string-based routing anywhere in the codebase.

6.3 State Hoisting & Unidirectional Data Flow

`ShoeCard` is a stateless composable that only receives `shoe: Shoe` and `onClick: () -> Unit`. Events flow up (`onClick` lambda), state flows down (data parameters).

6.4 Material Design 3

All screens use `Scaffold` with `TopAppBar`. `HomeScreen` includes a `NavigationBar` (bottom navigation) with Home, Favorites, Cart, and Profile tabs. All styling uses `MaterialTheme.colorScheme` and `MaterialTheme.typography`.

7 Features

7.1 Home Screen

Displays shoes in a 2-column responsive grid with:

- M3 `TopAppBar` with app title and search/cart icons
- M3 `NavigationBar` (bottom) with 4 tabs
- Product images loaded asynchronously from network
- Brand name, product name, and price
- Press animation feedback (scale to 96%)
- Cart badge showing item count

7.2 Detail Screen

Shows full product information:

- M3 `Scaffold` with `TopAppBar` (back + favorite icons)
- Large product image with background
- Brand, name, and description
- Size selector (7–11)
- Bottom bar with price and “Add to Bag” button

7.3 Cart Screen

Shopping bag management:

- M3 `Scaffold` with `TopAppBar` (back button + item count)
- Item cards with thumbnail, name, brand, price
- Quantity controls (increment/decrement)
- Per-item total calculation
- Subtotal, shipping (free), and grand total
- Checkout button

8 Screenshots

8.1 Home Screen

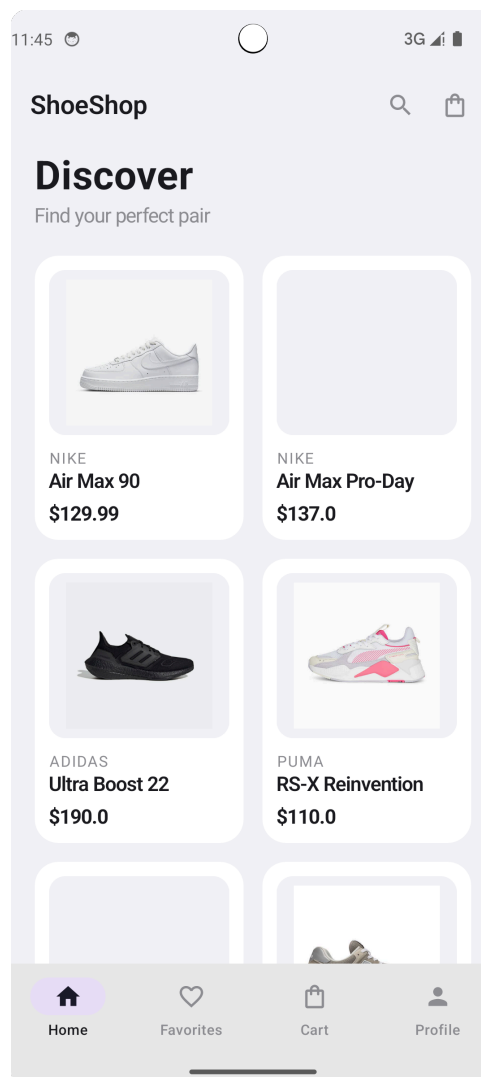


Figure 1: Product catalog with M3 TopAppBar and NavigationBar

8.2 Detail Screen

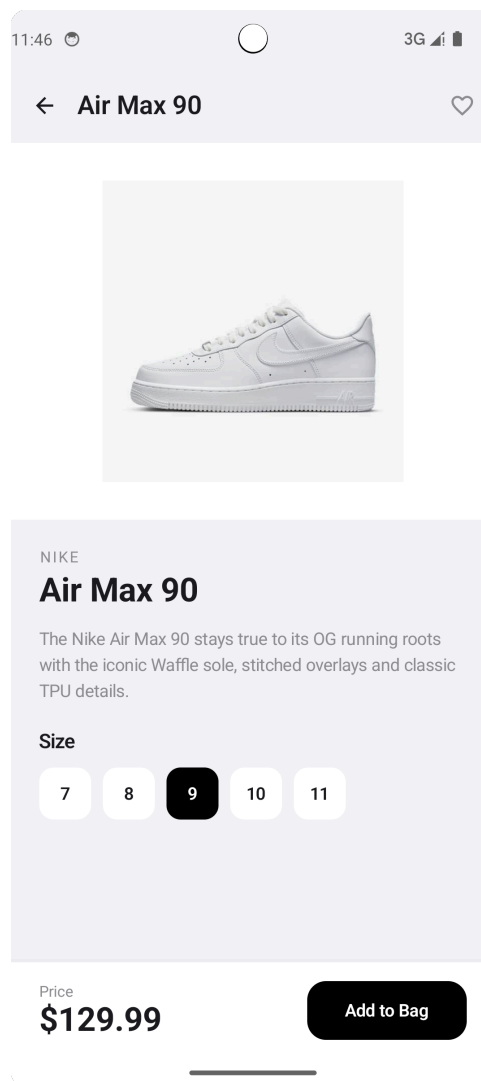


Figure 2: Product detail with Scaffold, TopAppBar, and Add to Bag

8.3 Shopping Cart

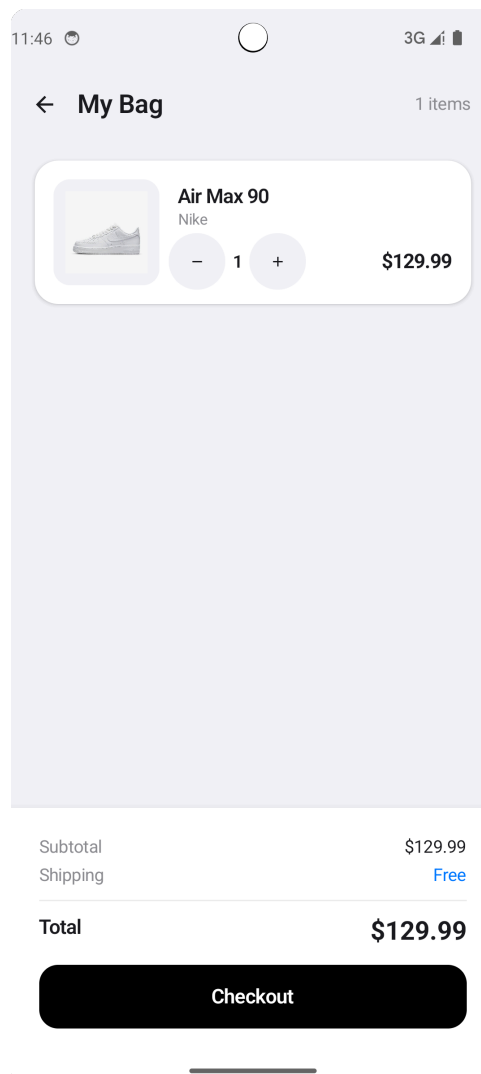


Figure 3: Shopping bag with items and total calculation

8.4 Cart with Multiple Items

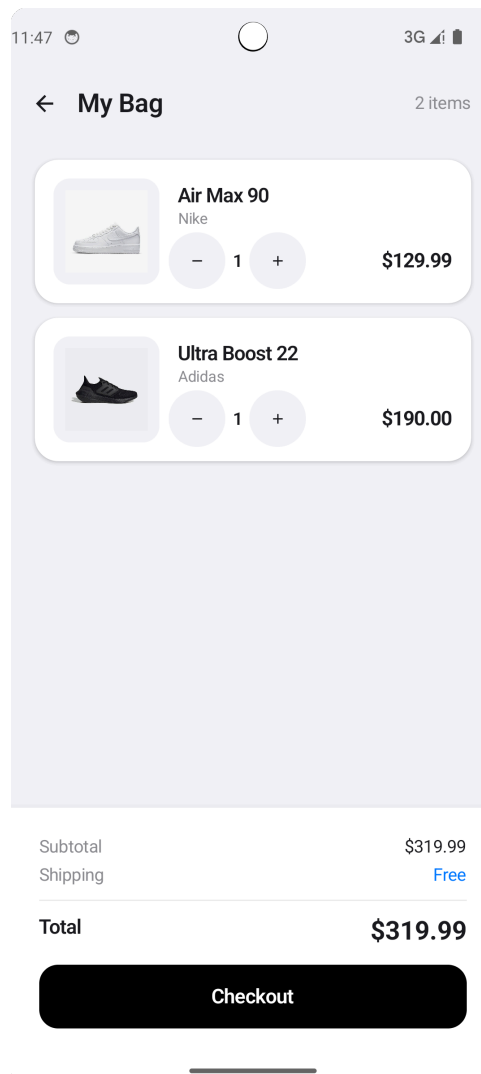


Figure 4: Shopping bag with multiple items and quantity adjustment